PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila











PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

CJ BIO

Histidine

Low CP Diets in the Philippines : The Emerging Role of Histidine

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Global Technical Head
CJ BIO

Contents

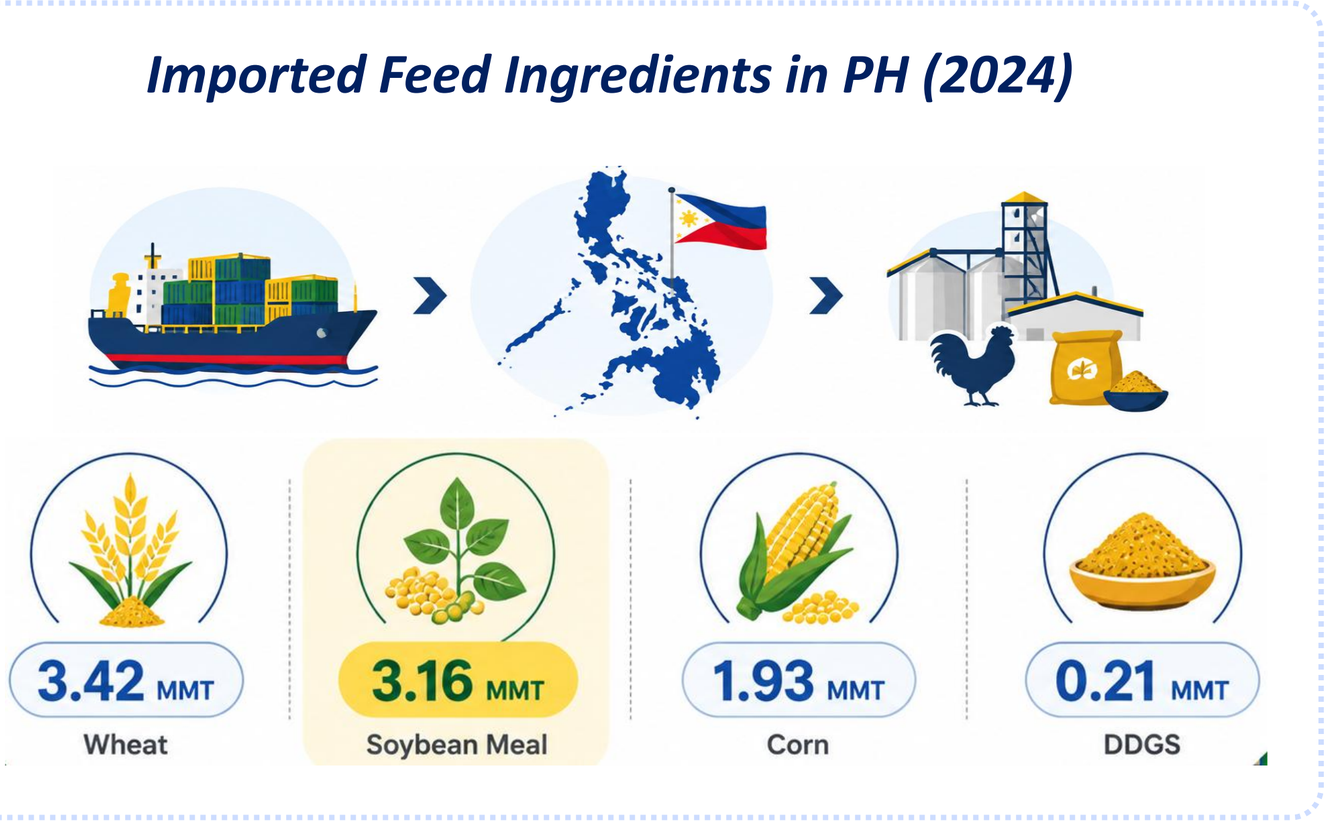
1. Protein Sources in the Philippines
   - ✓ Low CP and Histidine

2. Histidine Requirement
   - ✓ Recent Nutrient Guidance Updates
   - ✓ Requirement in Low CP Diets
   - ✓ Requirement under Pathogenic Challenges

3. Beyond Growth: Functional Roles of Histidine
   - ✓ Antioxidant Capacity and Carnosine

The Philippines Heavily Relies on Imported Protein Sources

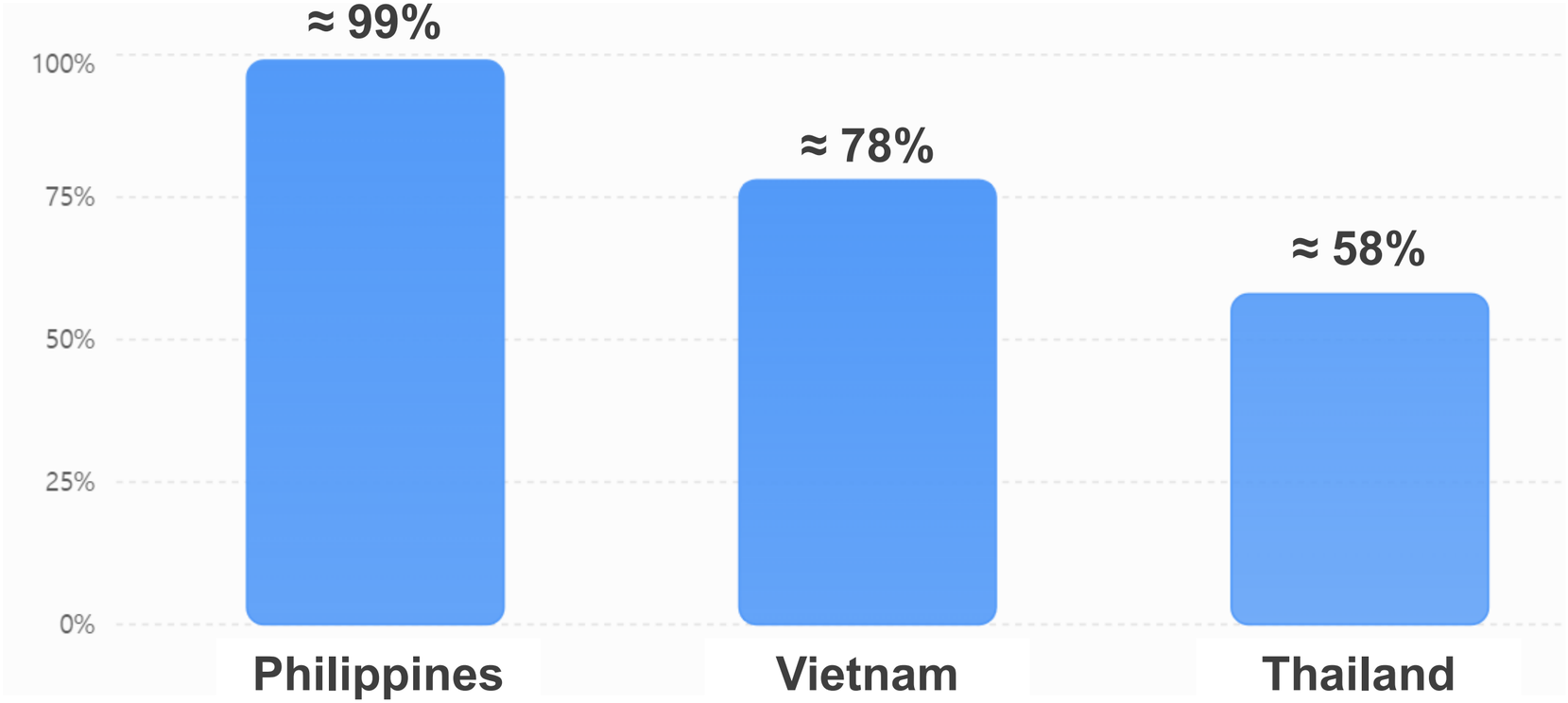
- Philippine protein supply is import-dependent (3.16 MMT soybean meal, 2024).
- This exposes producers to cost and supply volatility, driving the need for more efficient protein utilization.



Ref: USDA FAS Philippines: Oilseeds and Products Annual (2026)

How Does the Philippine Market Compare to its Neighbors?

- **Nearly 99% of the domestic soybean meal supply relies on direct imports**

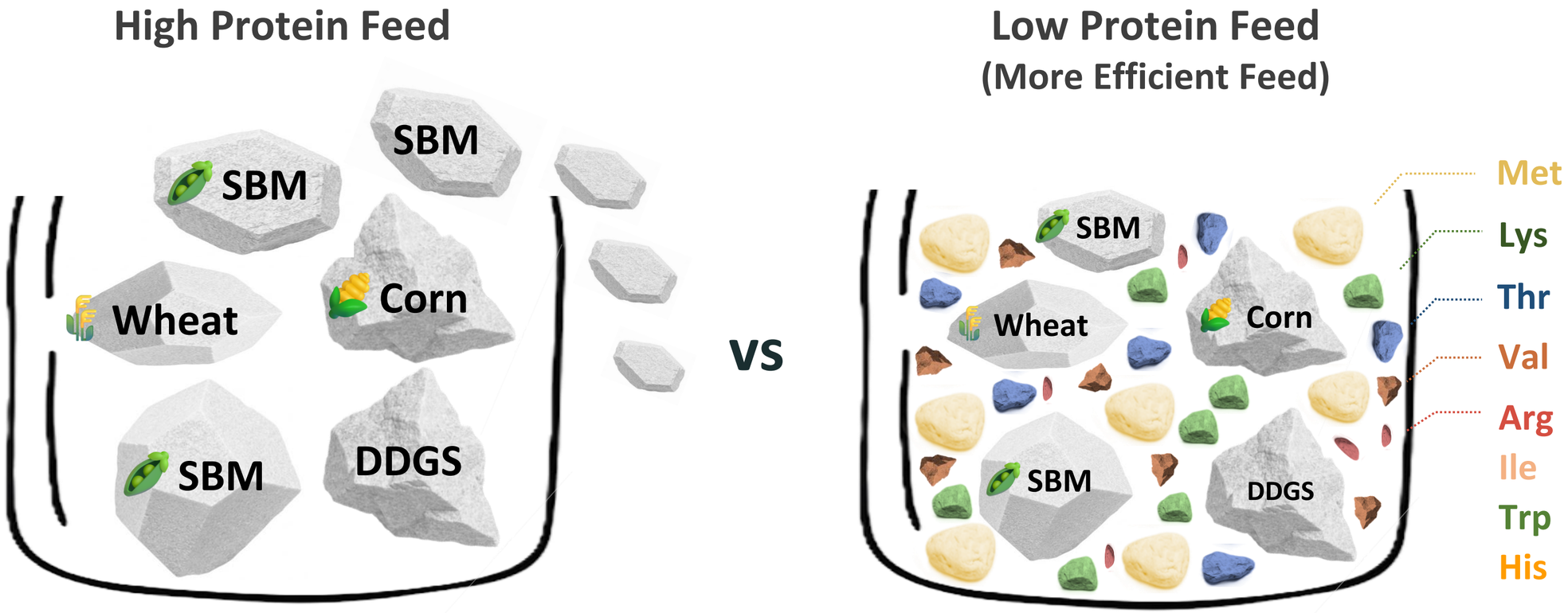


Share of Imported Soybean Meal in Domestic Consumption

Ref: USDA FAS Philippines: Oilseeds and Products Annual (2026)

Can We Reduce Protein Input Without Compromising Performance?

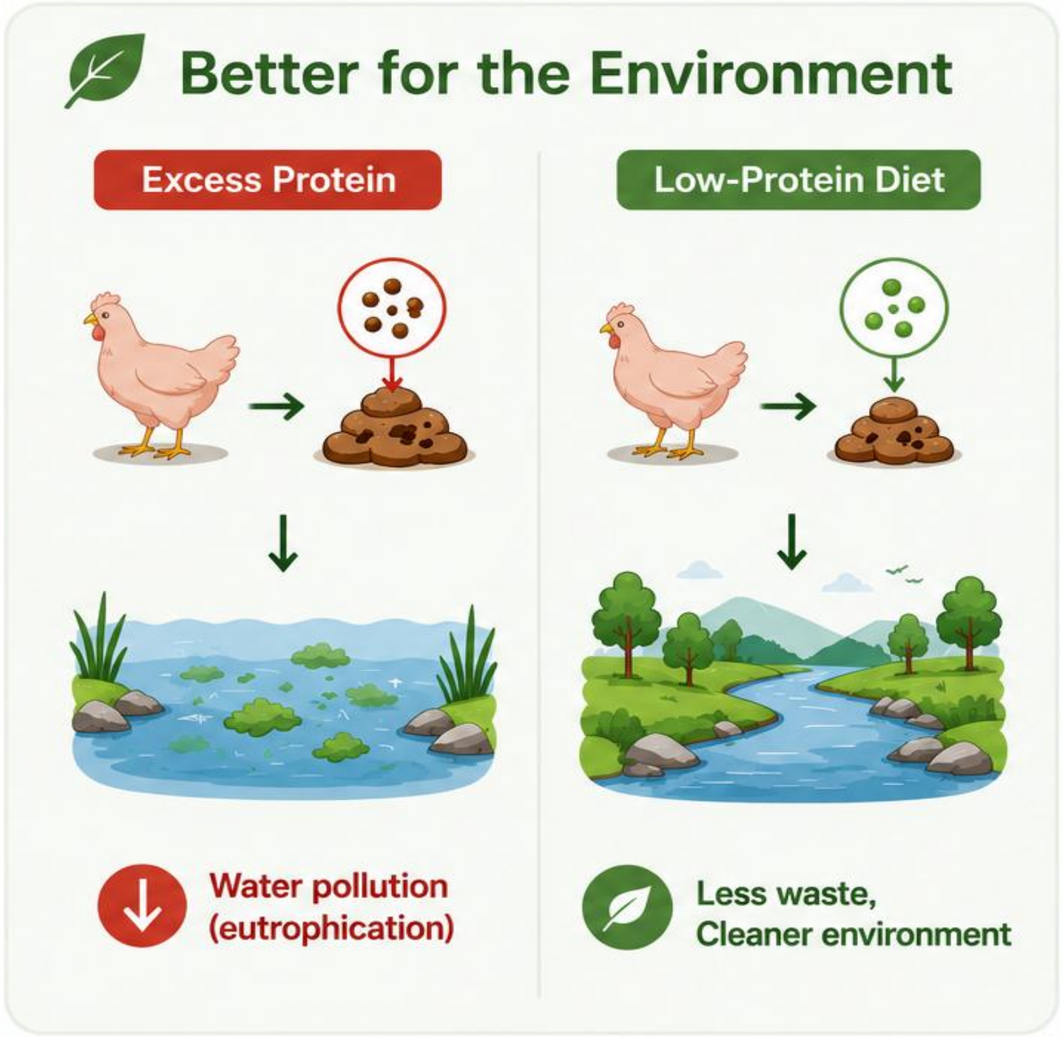
- Through low-CP precision nutrition, we can reduce intact protein while precisely balancing essential amino acids to maintain performance.

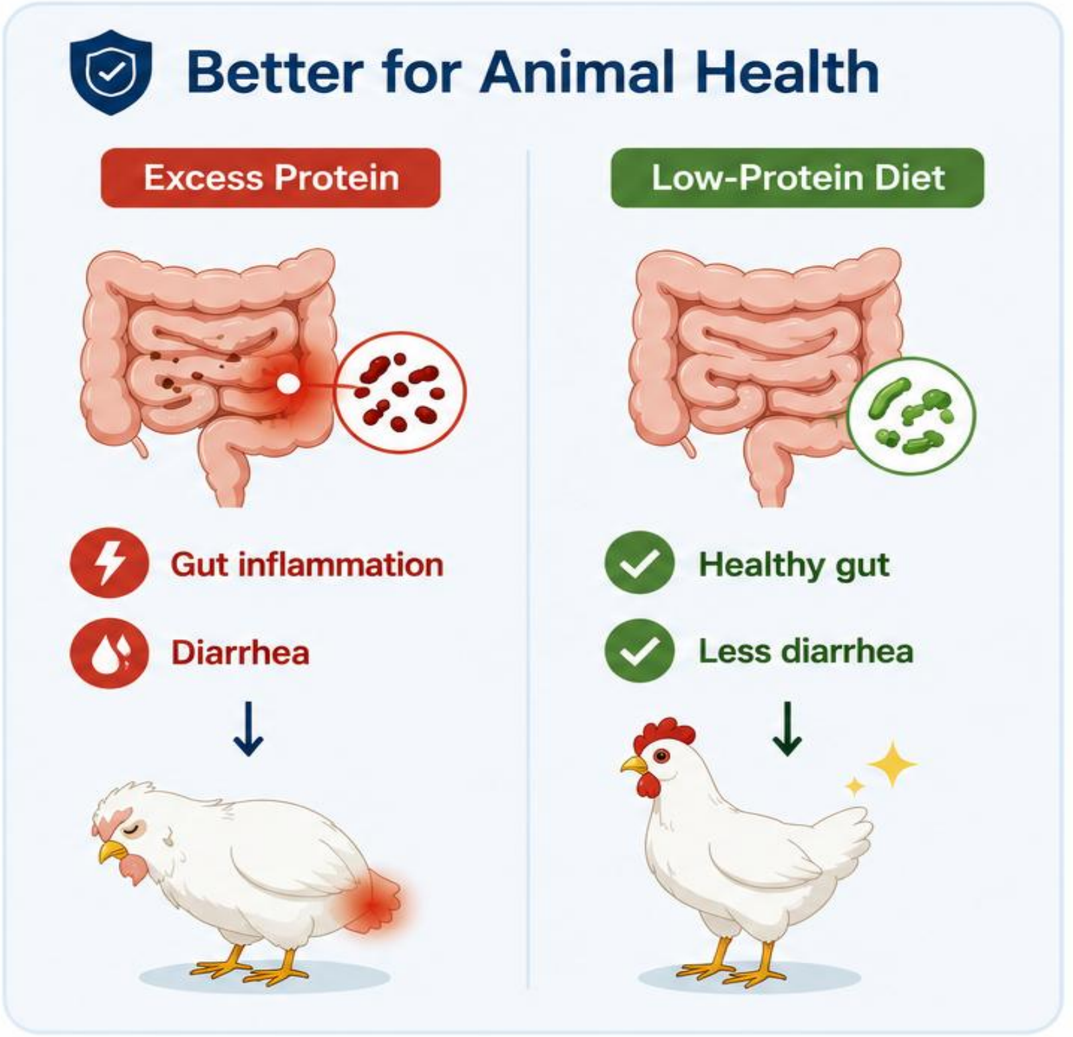


Low CP ≠ Less Nutrition — It Means More Precise Nutrition

Low CP Diets are Better for Environment and Health

- **Precisely meet animal needs → Less waste, healthier animals, and sustainable future**

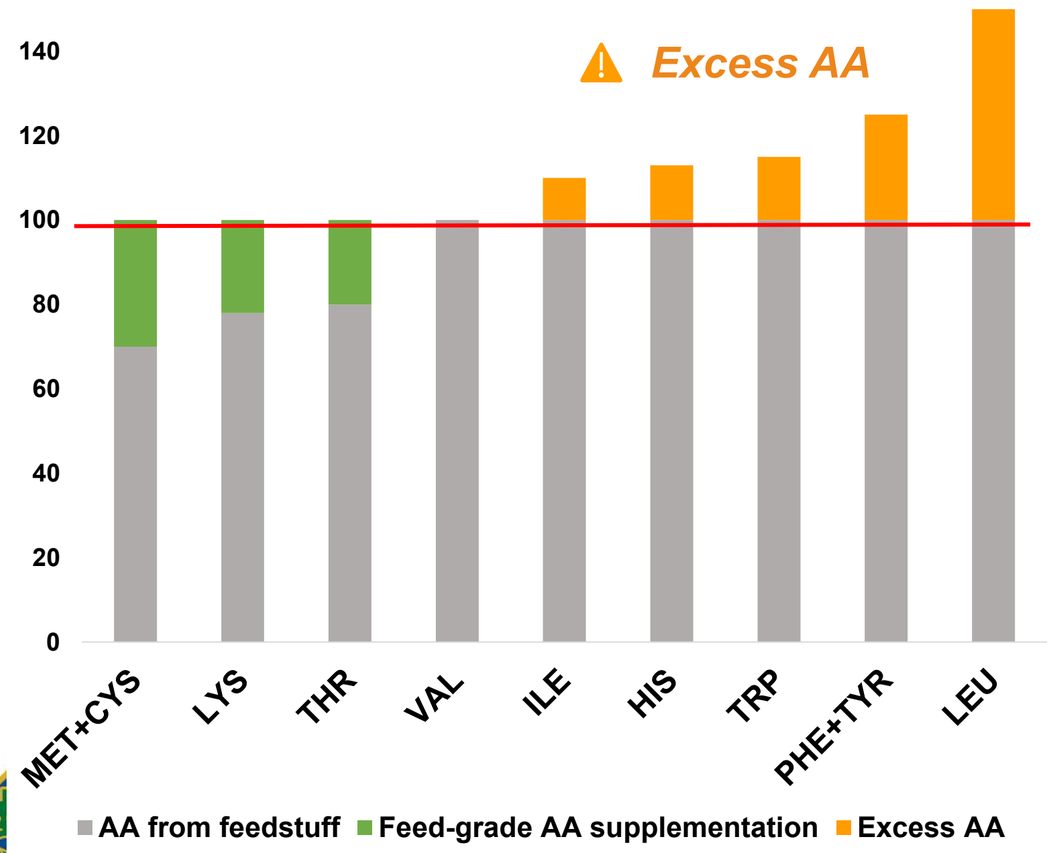


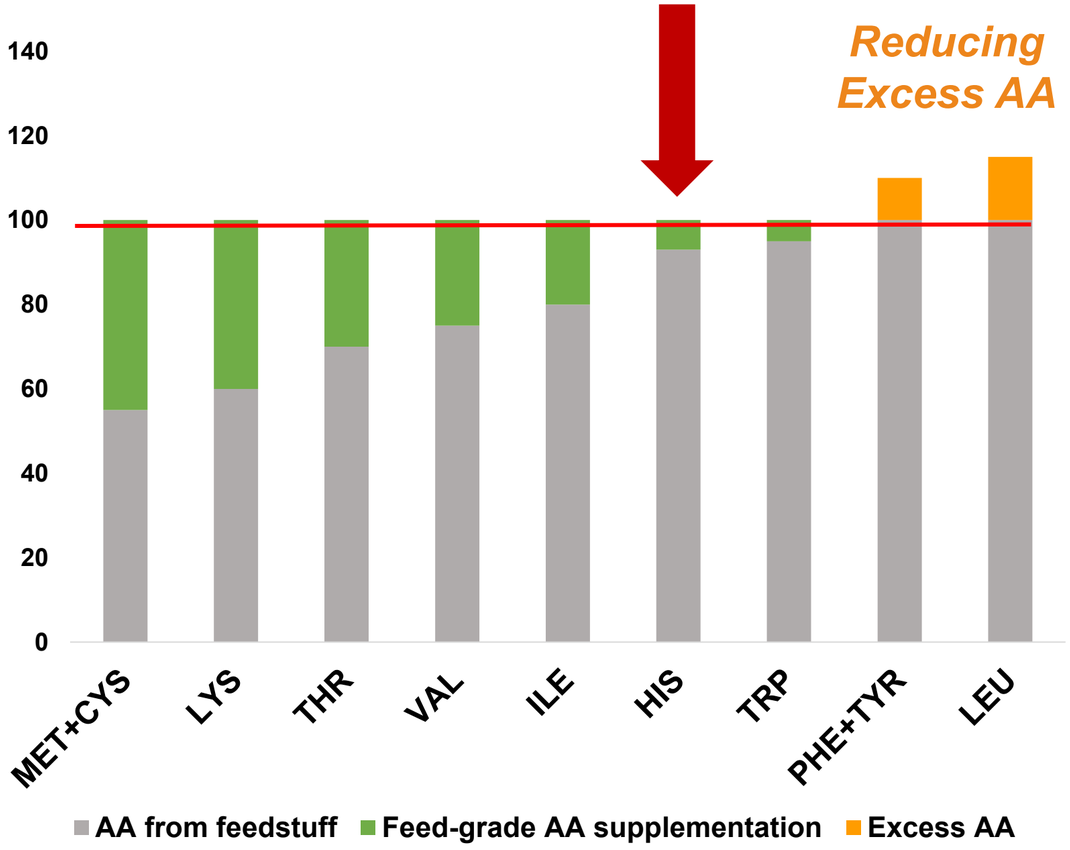


Protein vs. Amino Acids

Reducing CP & Balancing AA

Reducing N excretion





Histidine: An Underappreciated Amino Acid in Precision Poultry Nutrition

Histidine Requirement Is Being Re-evaluated

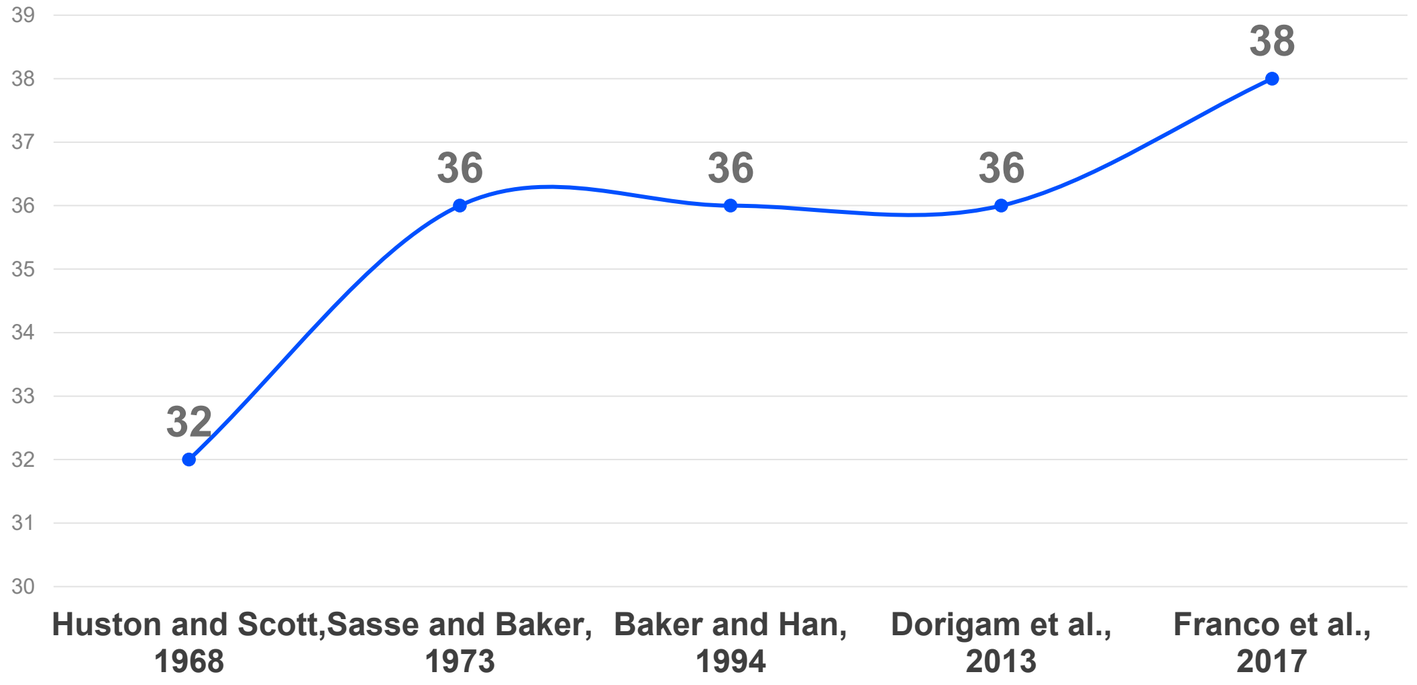
- Modern recommendations point toward a higher His:Lys ratio.
- The 2026 NRC revision also increased the recommended histidine requirement.

Nutrient Standard Guidance

SID His:Lys, %	1994	2017	2026
NRC	32	-	37
Brazilian Tables	-	37	-

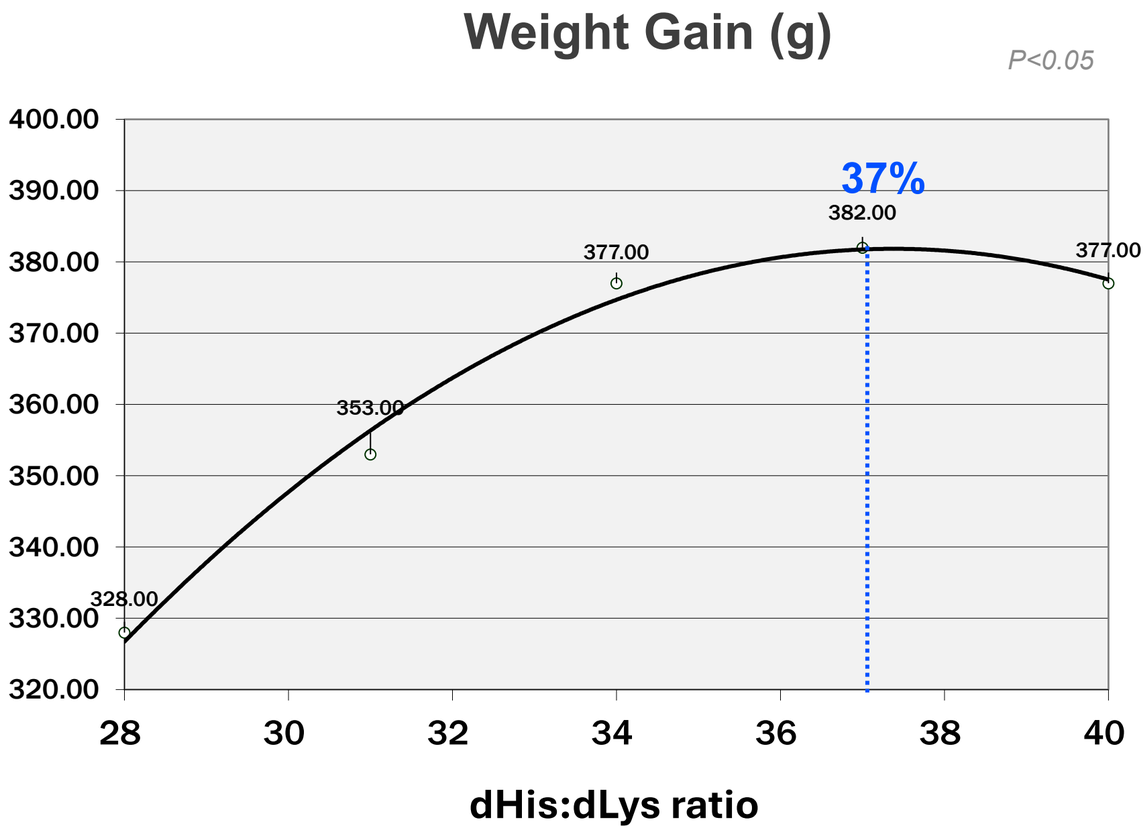
Recommended dHis:dLys Ratio Over Time

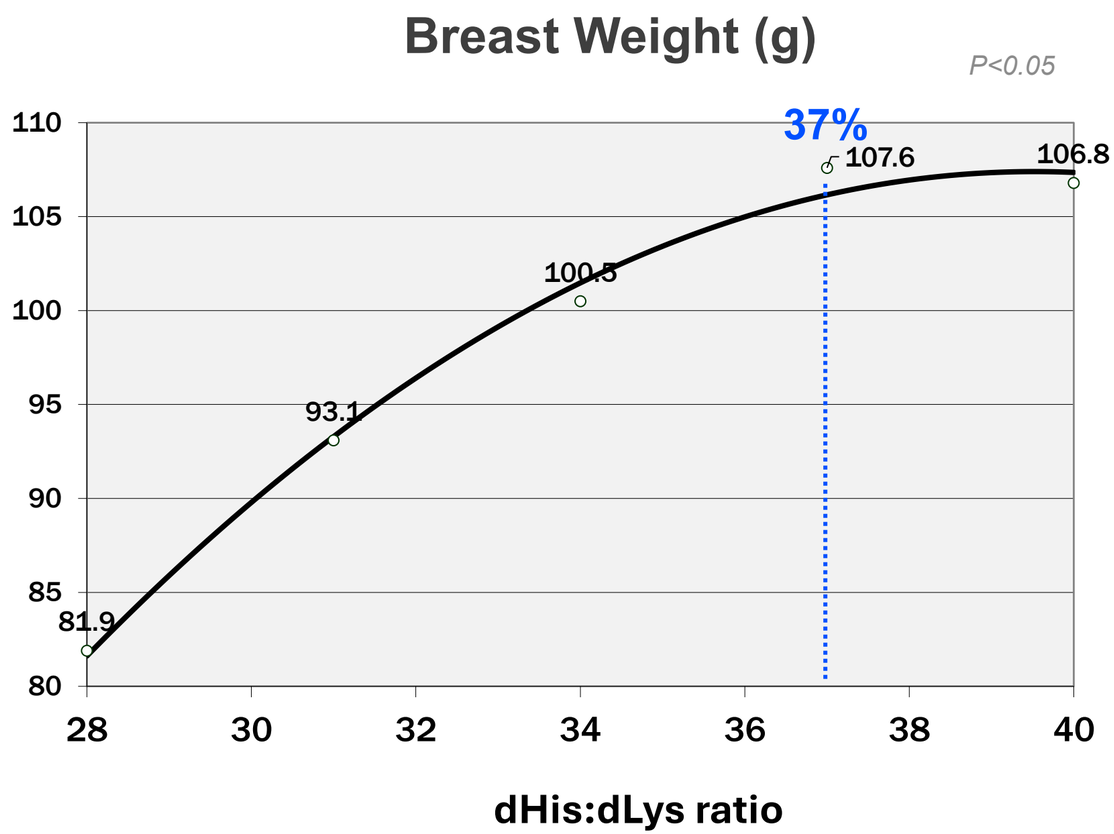
(Starter Phase)



Broilers Respond to Higher Histidine Levels (Cobb 500, Starter)

- **Supplementation of histidine increased the weight gain and quadratically decreased the FCR in Cobb 500 broilers from 8 to 17d**





Franco et al., 2017, Poultry Science

Broilers Need More HIS Than We Thought (Cobb 500, Starter)

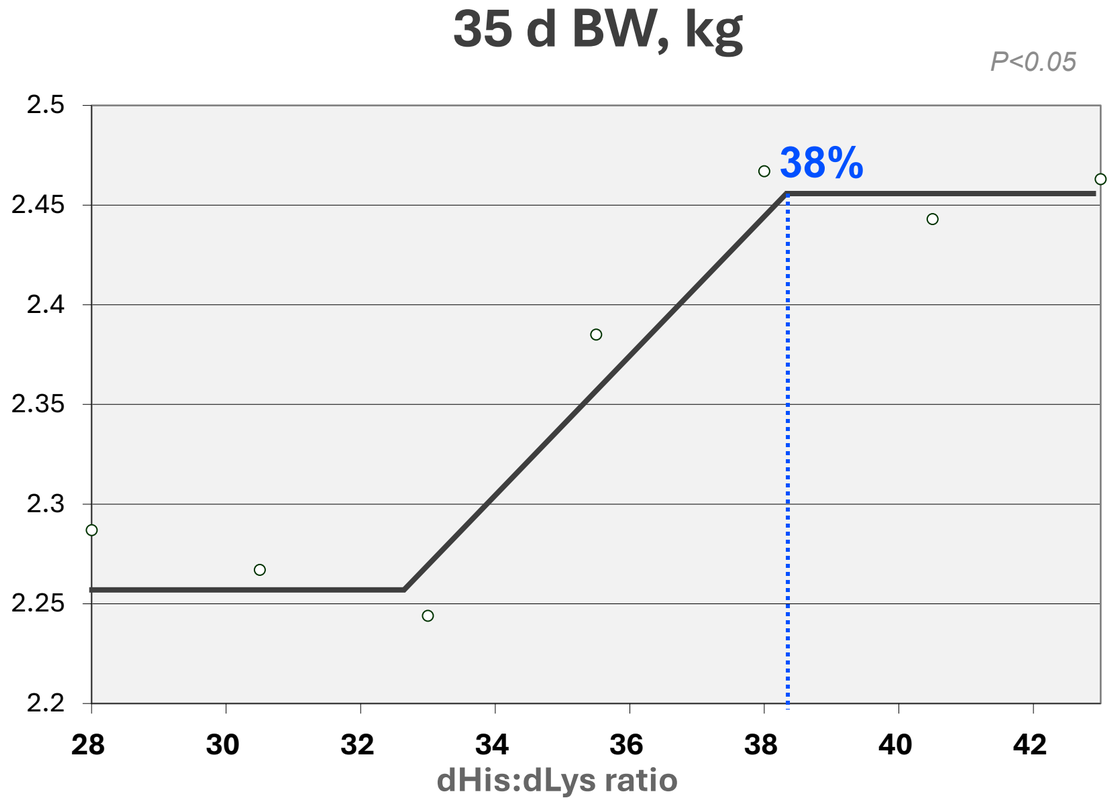
- **Based on growth performance and carcass parameters the estimated ideal dHis:dLys ratio is 38% for Cobb 500 broilers from 8 to 17d**

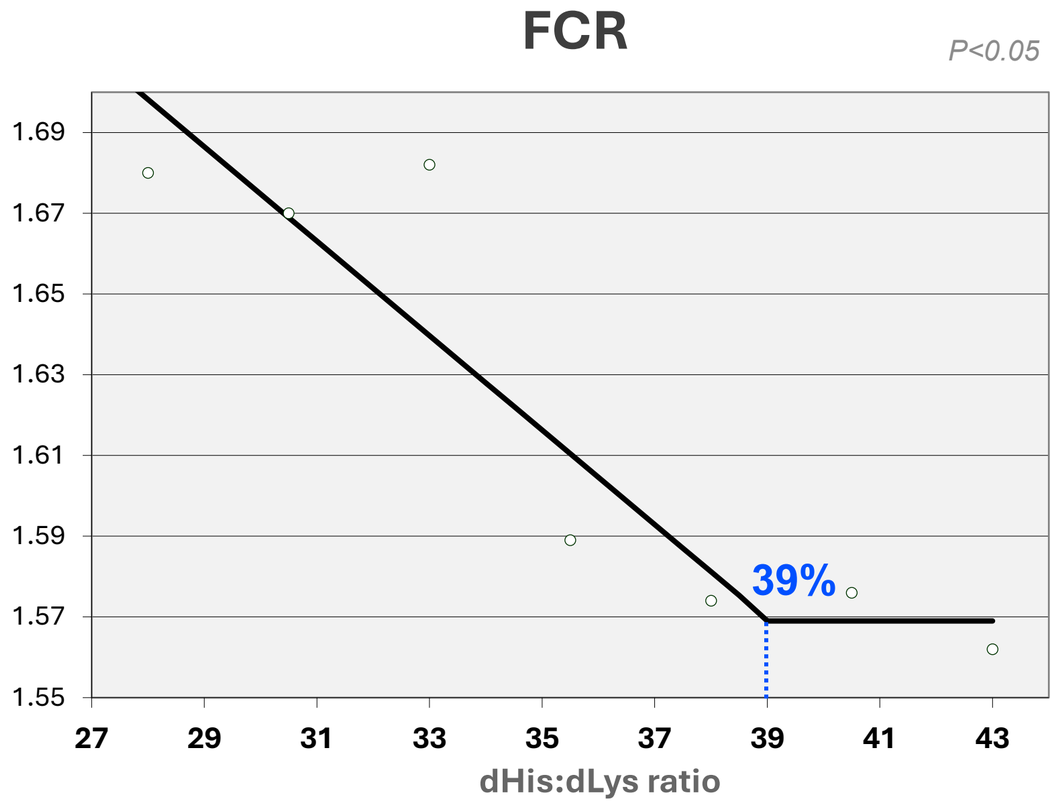
	SID His:Lys, %			
Response	Linear	Quadratic	95% Quad.	Mean ratio
Body weight gain	40	–	–	**38**
Feed conversion ratio	–	36	**34**	
Breast weight	–	39	**37**	
Breast fillet weight	–	41	**39**	
Breast yield	–	41	**39**	
Breast fillet yield	–	39	**40**	

Franco et al., 2017, Poultry Science

Histidine Requirements in Cobb under Low CP Diets

- **Diet CP level was 18.2%, male Cobb 500, d 15-35**
- **Under a low-CP diet (18.2% CP), increasing Histidine continuously improved growth performance.**

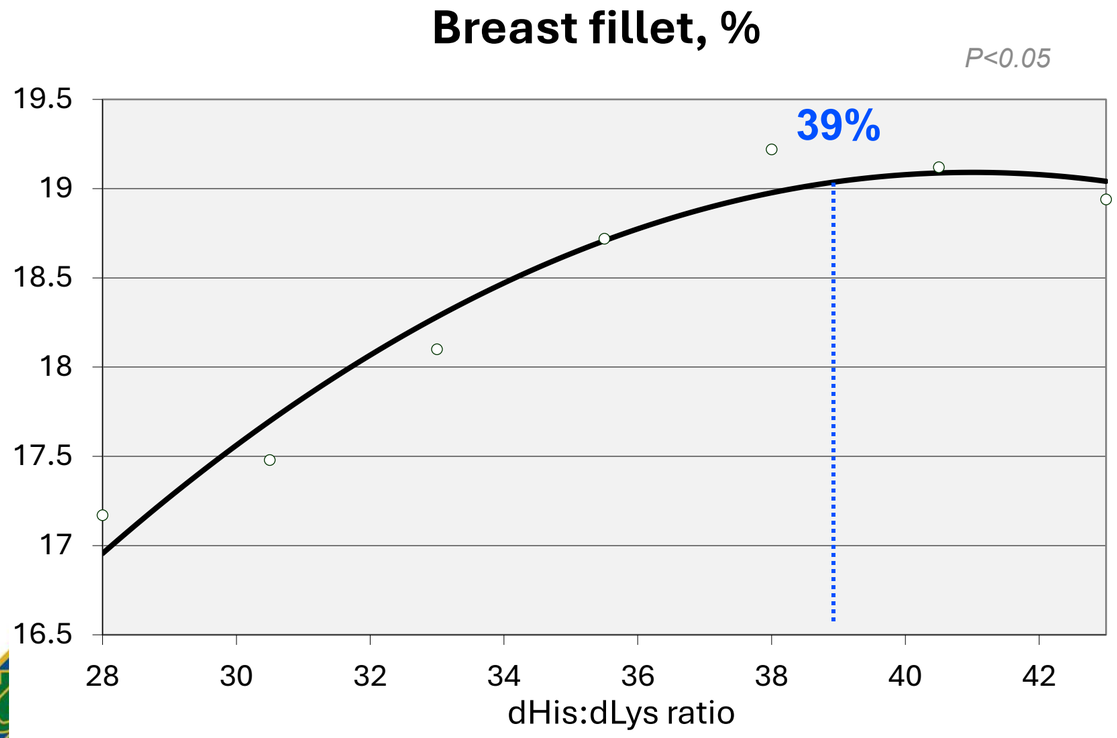


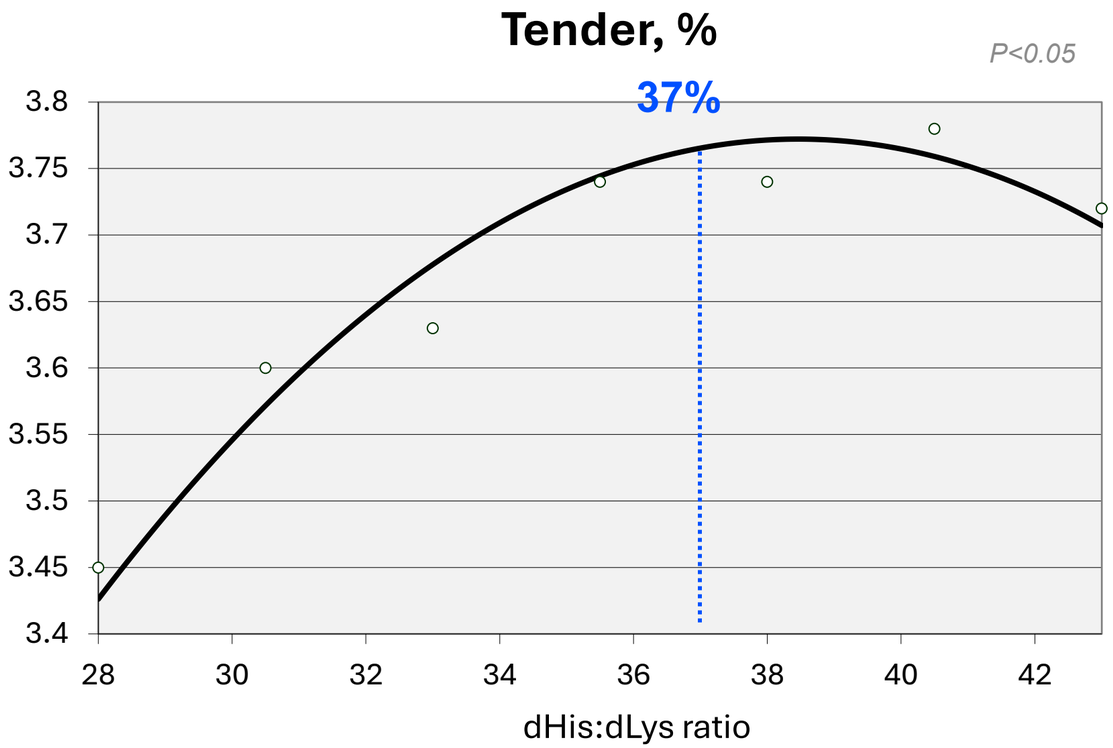


Nelson et al., 2024, J. Appl. Poult. Res, https://doi.org/10.1016/j.japr.2024.100431

Histidine Requirements in Cobb under Low CP Diets

- **Supplementing histidine led to improvement in the carcass performance of Cobb 500 broilers (d36 post-hatch).**
- **The respective dHis:dLys ratio of 39% and 37% are proposed to maintain breast fillet and tender yield for male Cobb 500 broilers**

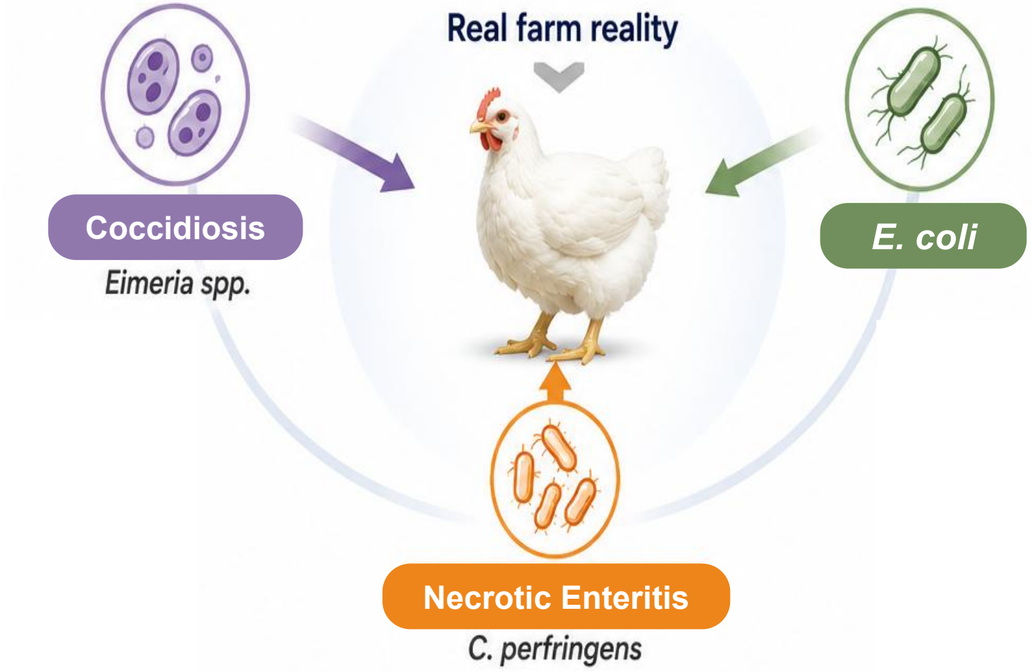


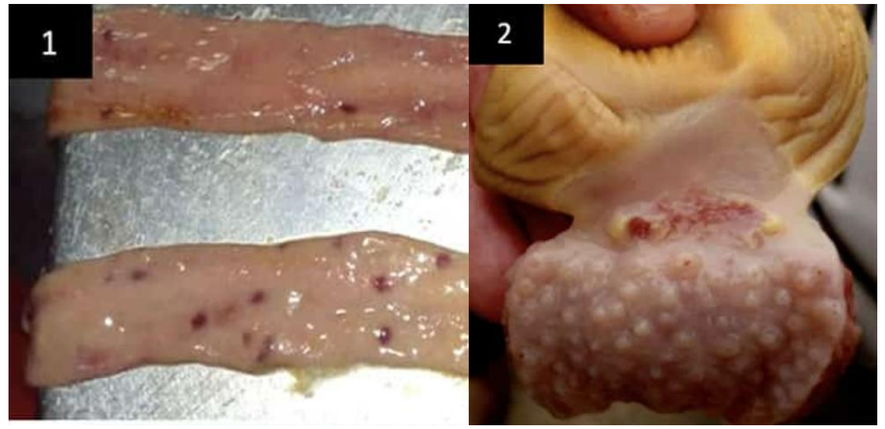


Nelson et al., 2024, J. Appl. Poult. Res, https://doi.org/10.1016/j.japr.2024.100431

Real-World Broilers Face Multiple Disease Challenges

- Under commercial conditions, broilers face coccidiosis and enteric bacterial infections simultaneously.
- These challenges damage the gut and activate immunity, altering histidine utilization.



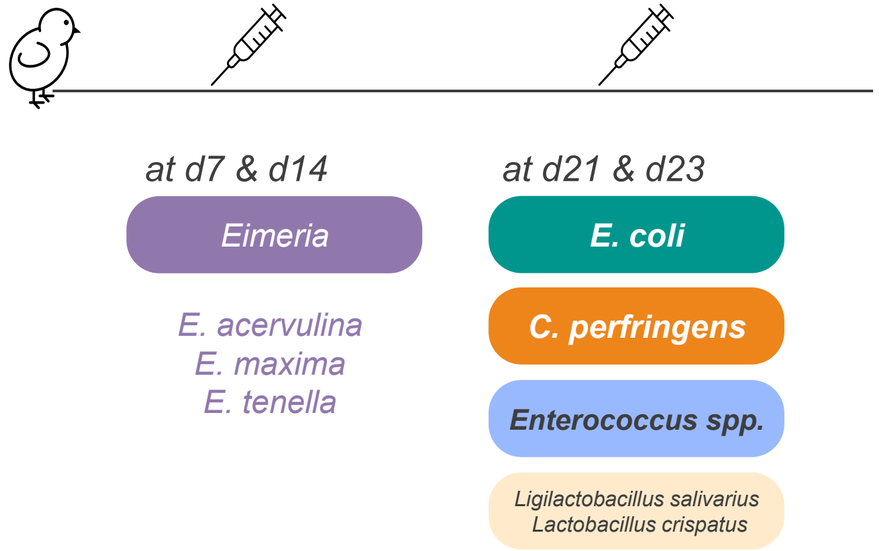


[Lesions in the (1) gut mucosa, (2) transit area between the gizzard and ventriculus caused by *C. perfringens*]

Verifying HIS requirement under Challenged Condition

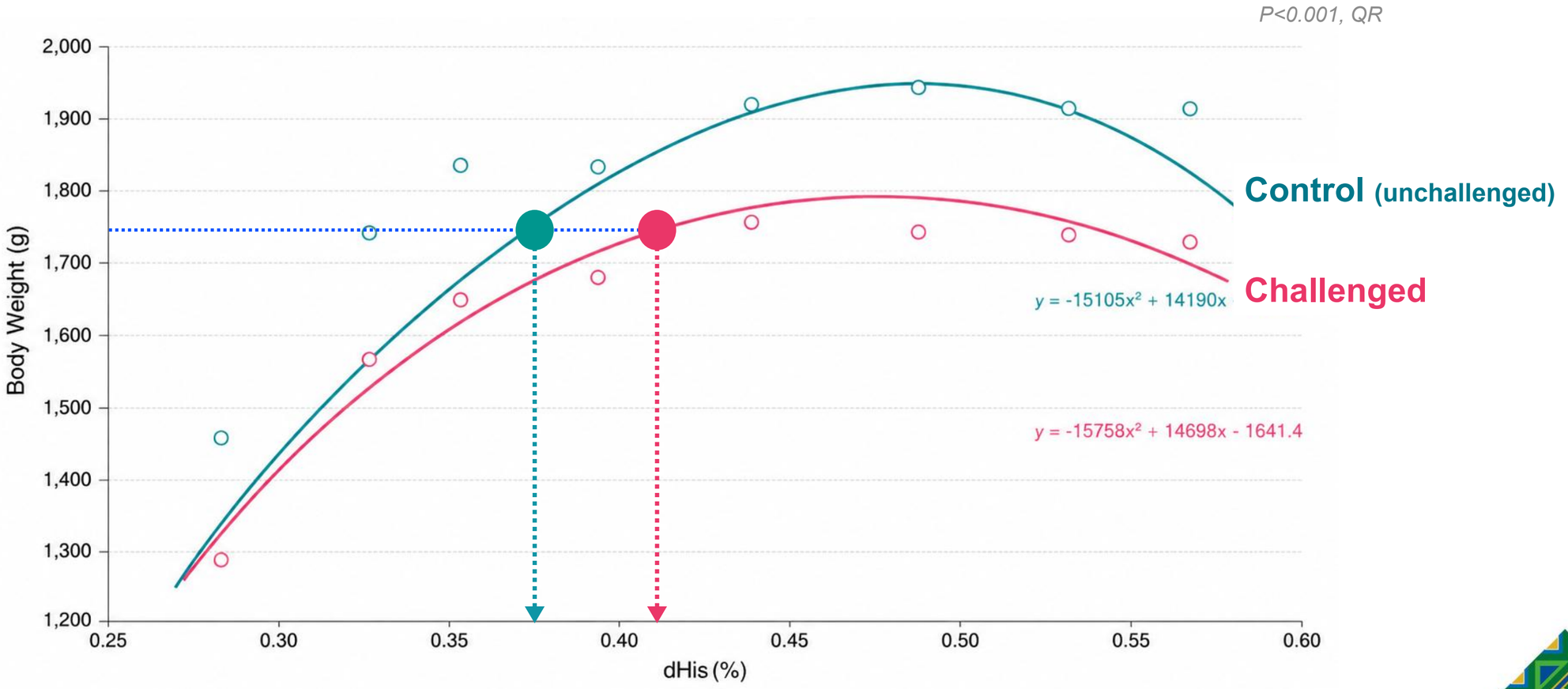
- His requirement was assessed in birds challenged with coccidial and enteric disease-inducing pathogens.
  - ✓ Coccidiosis challenged : a mixture *of Eimeria acervulina*, *Eimeria maxima* and *Eimeria tenella*
  - ✓ Bacterial enteritis challenged : a mixture of *E. coli, C. perfringens, L. salivarius, L. crispatus, and Enterococcus spp.*

TRT								
	Unchallenged	T01	T02	T03	T04	T05	T06	T07
	Challenged	T08	T09	T10	T11	T12	T13	T14
SID His level		0.30	0.33	0.36	0.39	0.44	0.49	0.54
His supplementation (g/mt)		0	300	600	900	1400	1900	2400



Challenged Broiler Needs More HIS for Growth

Body Weight (g) – Challenged vs Unchallenged Ross 308 (D28)



Challenged Broiler Needs More HIS

- Achieving an equivalent growth performance and carcass yield target required a higher histidine supply under challenged conditions.

HIS Requirement for same performance target (98% of C-group maximum)

Parameter	Unchallenged (% SID HIS)	Challenged (% SID HIS)	Increase (delta)
Body Weight (d35)	0.387	0.407	+5.2%
FCR	0.325	0.375	+15.4%
Slaughter Yield	0.428	0.472	+10.3%
Live Weight	0.359	0.381	+6.1%
Carcass Weight	0.369	0.408	+10.6%
Breast Weight	0.413	0.490	+18.6%

Challenged Broiler Needs More HIS

1. **Histidine deficiency clearly compromises broiler performance.**
2. **Immune challenge increases the SID His requirement, particularly during the acute challenge period.**
3. **Histidine requirement for breast meat production may be higher than that required for growth and FCR.**

- Under challenge, the His requirement for maximum performance at day 35 is:
  - **0.37–0.41% SID His** for growth performance parameters
  - **0.38–0.49% SID His** for slaughter yield, and breast yield/carcass

Under challenged situation, the Histidine requirement increases compared to the unchallenged condition in order to achieve optimal growth performance and carcass yield

Reduction of SBM or Use of Alternative Proteins meals

- ***Male Ross 308 birds (n = 1350)***
- ***Three-phase feeding experiment (0–12 d, 13–21 d, and 22–35 d)***

Treatment	T1	T2	T3	T4	T5	T6
Starter	21%	21%	21%	20%	21%	21%
Grower	20%	20%	19%	19%	20%	20%
Finisher	19%	17%	17%	17%	19%	19%
SBM					<15%	0%

Matton et al. Animals 2025, 15, 1753.
Saremi et al. 2023. Proceeding of European Symposium of Poultry Nutrition, Rimini, Italy.

Reduction of SBM or Use of Alternative Proteins meals

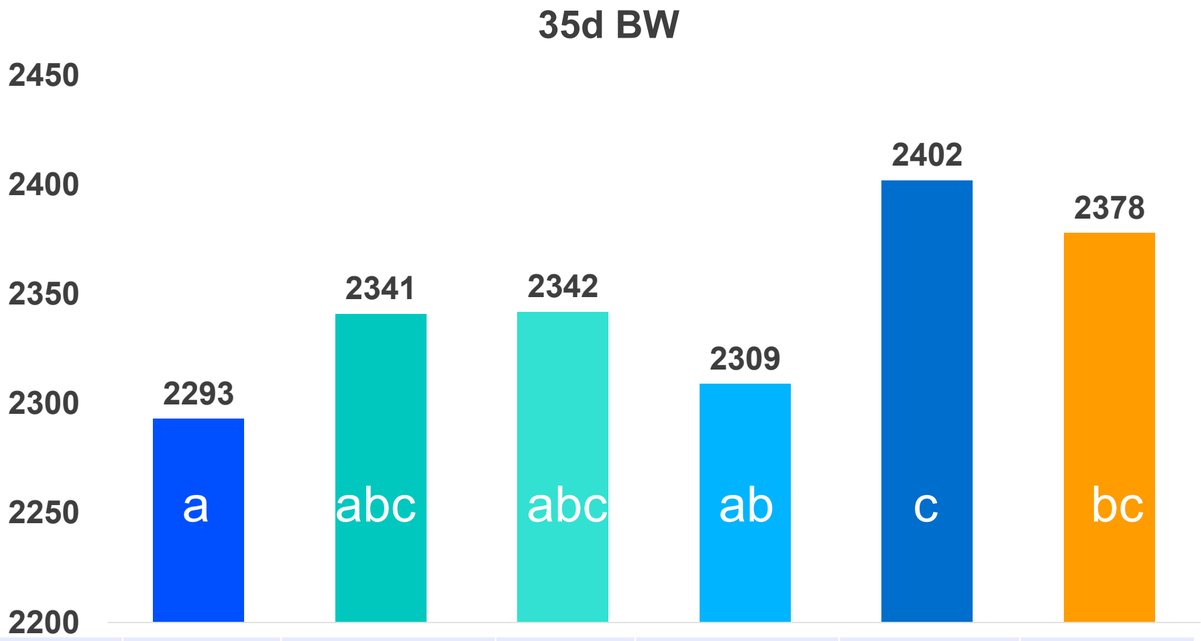
	Starter			
Ingredients (%)	T01 – T02 – T03	T04	T05	T06
Corn	45.56	49.93	36.80	38.38
Soybean meal	30.08	25.92	15.00	
Wheat	15.00	15.00	15.00	15.00
Soybean oil	3.35	2.56	4.07	2.23
Rapeseed meal			8.00	10.84
Field beans			7.00	8.00
Peas			4.61	8.00
Corn gluten meal			3.40	10.58
Others (Premix. MCP.NaCl, etc.)	4.54	4.60	4.24	4.31
L-Lysine HCl	0.33	0.47	0.5	0.83
L-Methionine 100	0.40	0.43	0.39	0.36
L-Threonine	0.25	0.30	0.26	0.30
L-Valine	0.14	0.21	0.18	0.24
L-Isoleucine	0.11	0.18	0.18	0.26
L-Arginine	0.20	0.32	0.29	0.51
L-Histidine HCL	0.04	0.07	0.05	0.08
L-Tryptophan		0.01	0.03	0.08
Calculated nutrients				
AMEn (kcal/kg)	3025.00	3025.00	3025.00	3025.00
Crude protein (%)	21.00	20.00	21.00	21.00
SID Lys (%)	1.25	1.25	1.25	1.25
SID His (%)	**0.48 (38%)**	**0.48 (38%)**	**0.48 (38%)**	**0.48 (38%)**

	Grower			
Ingredients (%)	T01 – T02	T03-T04	T05	T06
Corn	41.48	45.64	31.61	34.24
Soybean meal	28.67	24.76	15.00	
Wheat	20.00	20.00	20.00	20.00
Soybean oil	4.65	3.9	5.00	3.61
Rapeseed meal			4.14	10.59
Field beans			8.00	8.00
Peas			8.00	8.00
Corn gluten meal			3.17	9.79
Others (Premix. MCP.NaCl, etc	4.27	4.33	3.80	3.76
L-Lysine HCl	0.21	0.33	0.34	0.67
L-Methionine 100	0.33	0.36	0.34	0.29
L-Threonine	0.17	0.22	0.20	0.22
L-Valine	0.09	0.15	0.14	0.18
L-Isoleucine	0.04	0.11	0.10	0.18
L-Arginine	0.09	0.20	0.13	0.37
L-Histidine HCL			0.02	0.04
L-Tryptophan			0.01	0.06
Calculated nutrients				
AMEn (kcal/kg)	3107.00	3107.00	3107.00	3107.00
Crude protein (%)	20.00	19.00	20.00	20.00
SID Lys (%)	1.12	1.12	1.12	1.12

02 Histidine requirement

Reduction of SBM or Use of Alternative Proteins meals

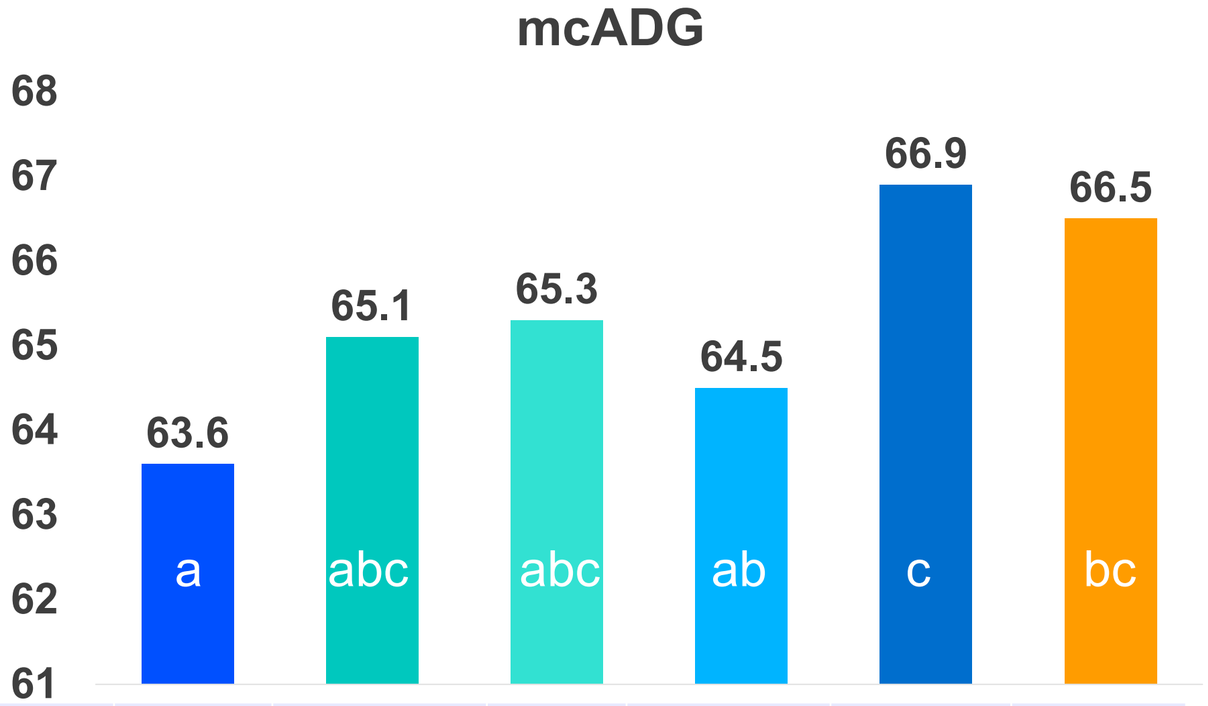
	Finisher			
Ingredients (%)	T01	T02-T03-T04	T05	T06
Corn	39.39	47.93	30.76	29.39
Soybean meal	25.86	17.64	15.00	
Wheat	25.00	25.00	25.00	25.00
Soybean oil	5.00	3.46	5.00	5.00
Rapeseed meal			1.05	12.39
Field beans			8.00	8.00
Peas			8.00	8.00
Corn gluten meal			2.67	7.37
Others (Premix. MCP. NaCl, et	3.95	4.15	3.43	3.23
L-Lysine HCl	0.19	0.46	0.29	0.55
L-Methionine 100	0.29	0.36	0.32	0.25
L-Threonine	0.15	0.25	0.18	0.19
L-Valine	0.06	0.19	0.10	0.13
L-Isoleucine	0.05	0.19	0.10	0.17
L-Arginine	0.06	0.30	0.08	0.27
L-Histidine HCL		0.06	0.01	0.02
L-Tryptophan		0.01	0.01	0.04
Calculated nutrients				
AMEn (kcal/kg)	3155.00	3155.00	3155.00	3155.00
Crude protein (%)	19.00	17.00	19.00	19.00
SID Lys (%)	1.04	1.04	1.04	1.04



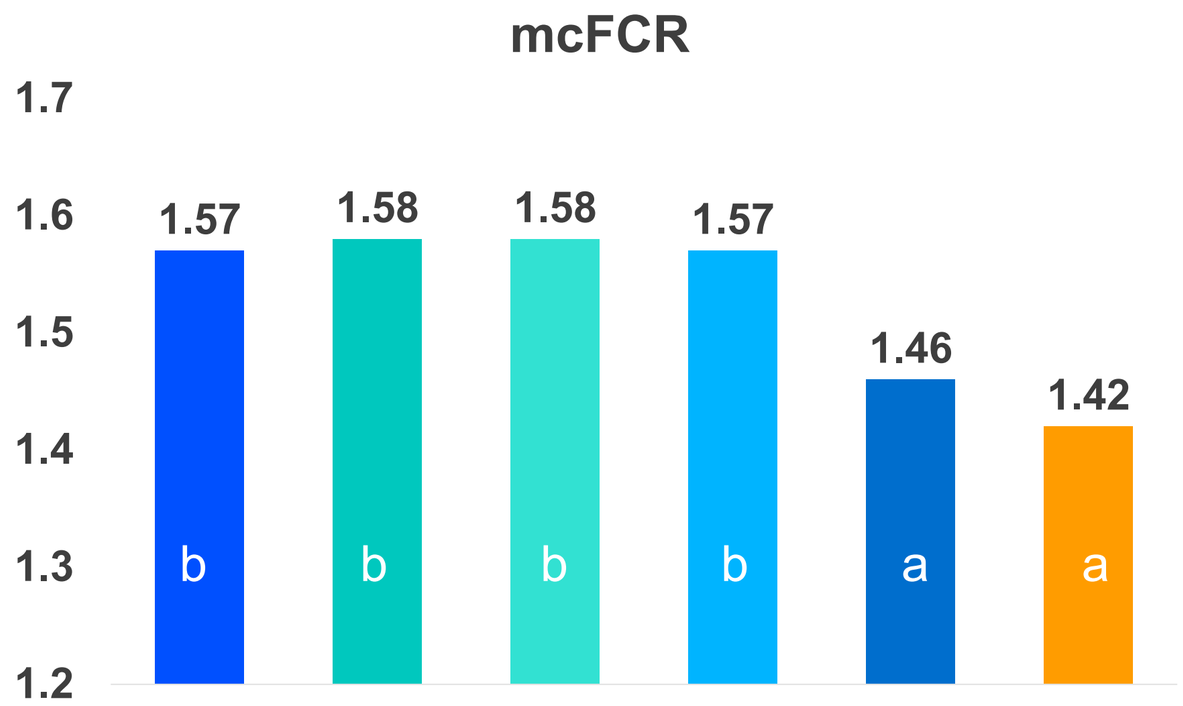
CP	T1	T2	T3	T4	T5	T6
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SBM					<15%	0%

Matton et al. Animals 2025, 15, 1753.

Reduction of SBM or Use of Alternative Proteins meals



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Matton et al. Animals 2025, 15, 1753.

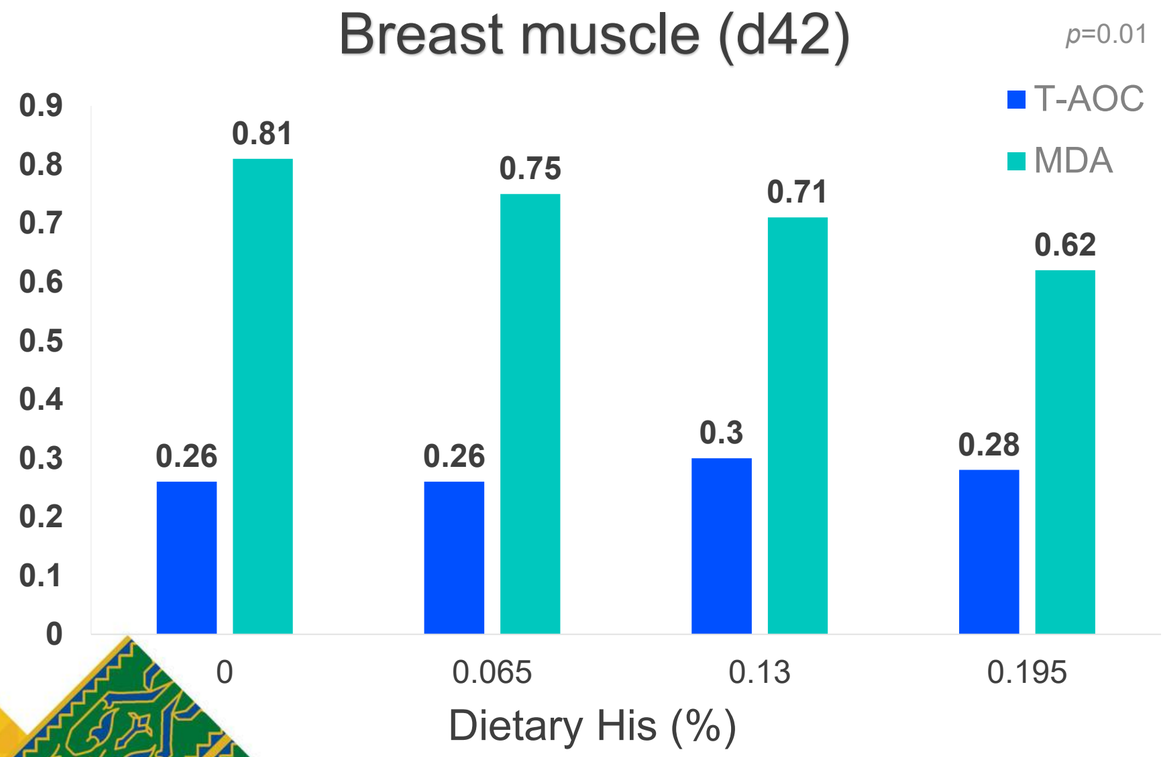
Multifunctional Role of Histidine

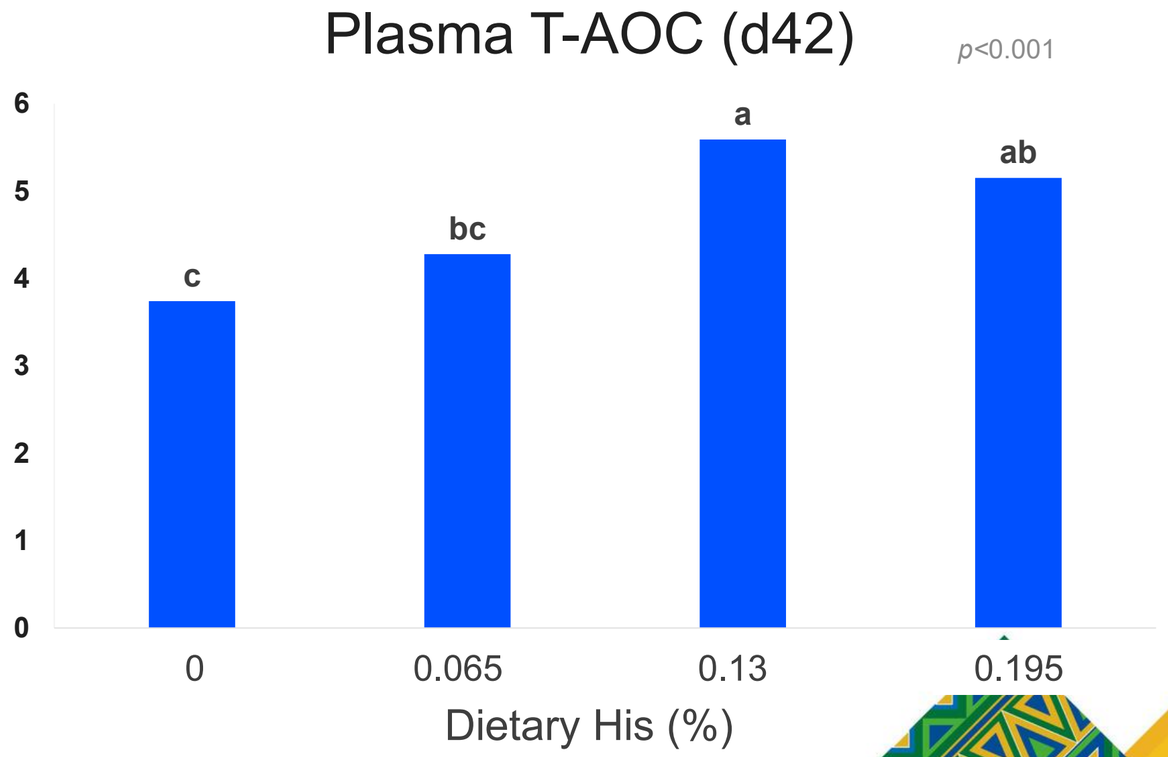
- Antioxidant
- Synthesis in Breast Muscle
- Hatch performance

HIS Increases Antioxidant Capacity

- Dietary addition of L-histidine could improve meat quality by increasing T-AOC (Total Antioxidant Capacity) in breast muscle and decreasing MDA level in plasma

640 male Arbor Acres male broilers, 1-42 d





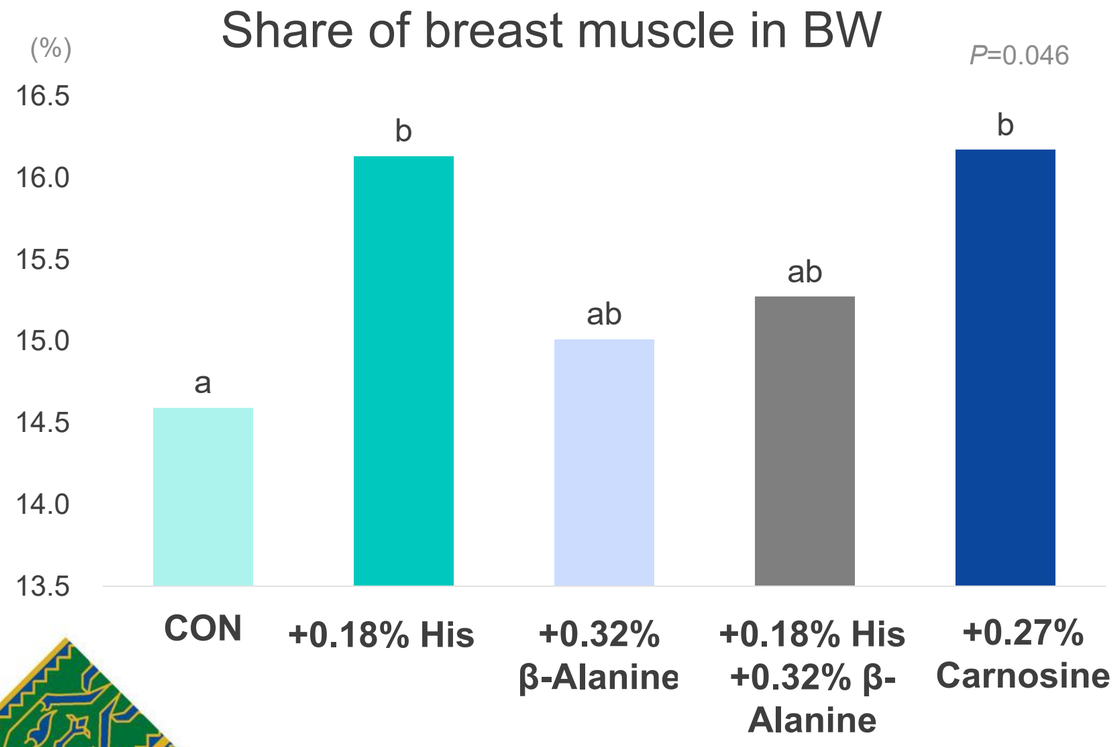
Qi et al., 2021, Animals

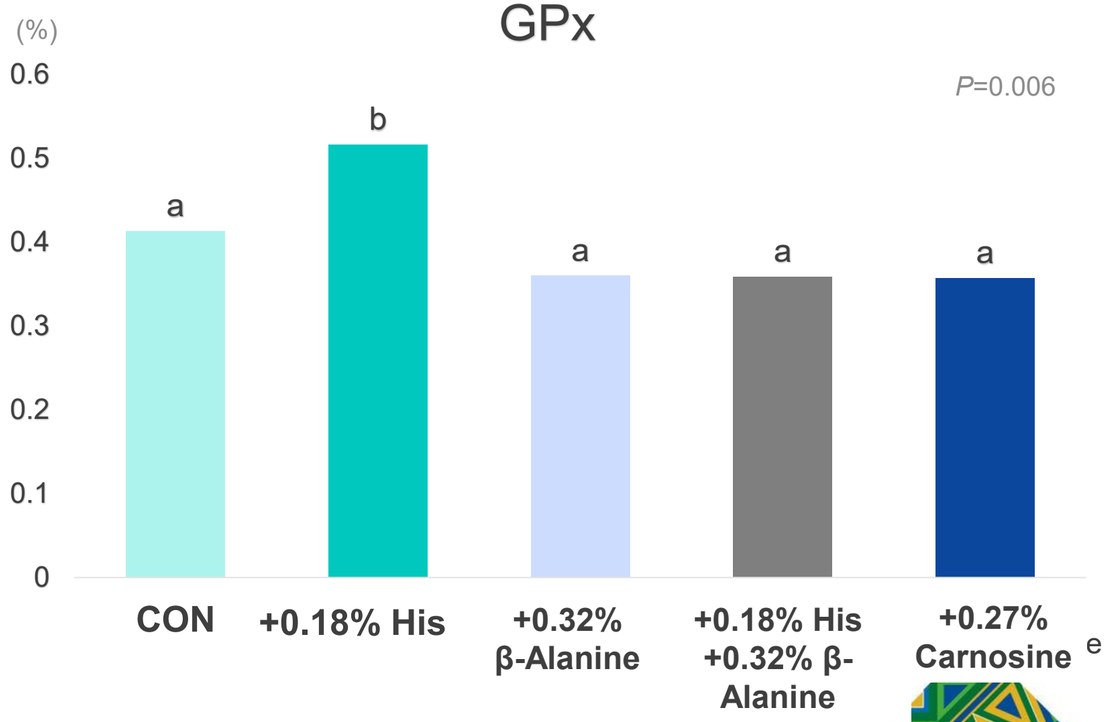
MDA: malondialdehyde

HIS Enhances Antioxidant Defense

- Dietary Histidine significantly increased GPx(Glutathione Peroxidase) activity in broilers.
- His and carnosine supplementation increase share of breast muscle

Hubbard Flex male broilers (1-28 d)



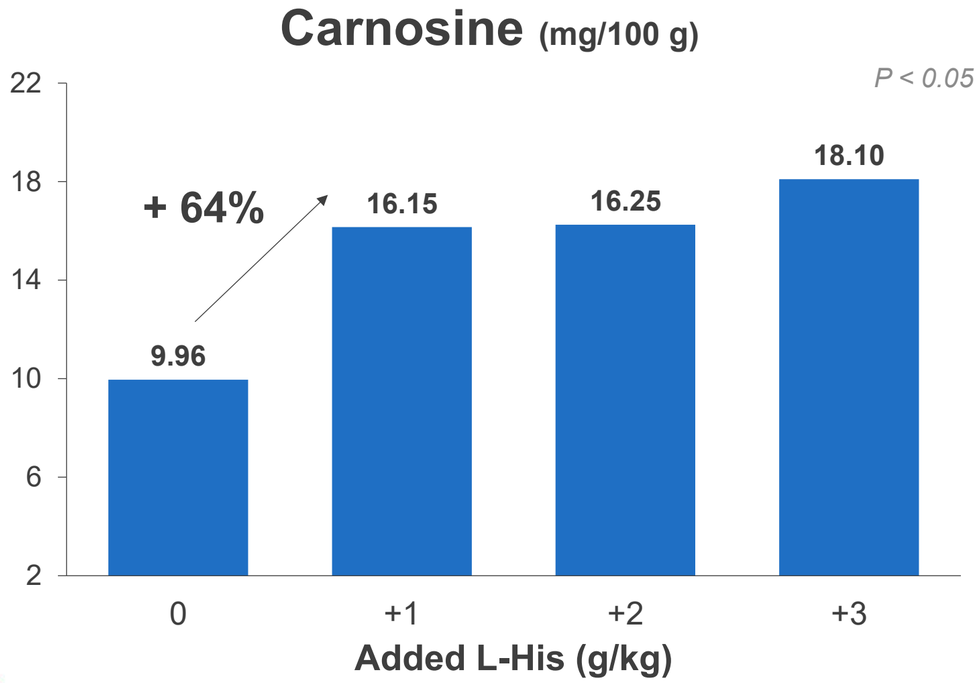


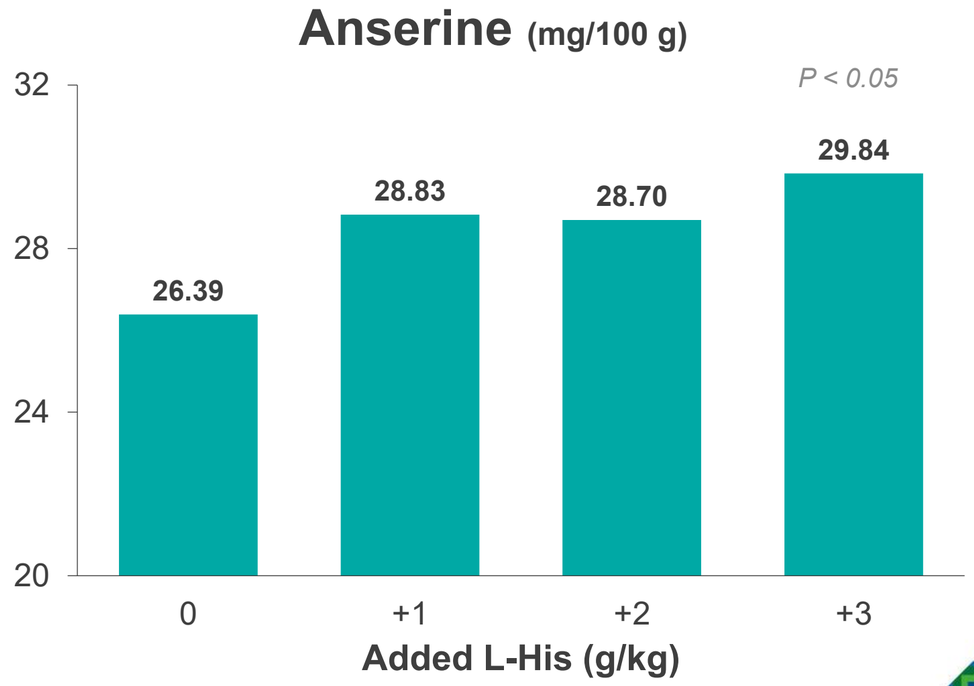
Kopec et al., 2020, Antioxidants

Histidine Drives Carnosine Synthesis in Breast Muscle

- Dietary Histidine markedly increased Carnosine and Anserine concentrations in breast muscle.
  - ✓ Histidine + β-Alanine → Carnosine → pH Buffering · Antioxidant Capacity

Male Ross 308 | d10–28 | Commercial diet + 0–3 g/kg L-His



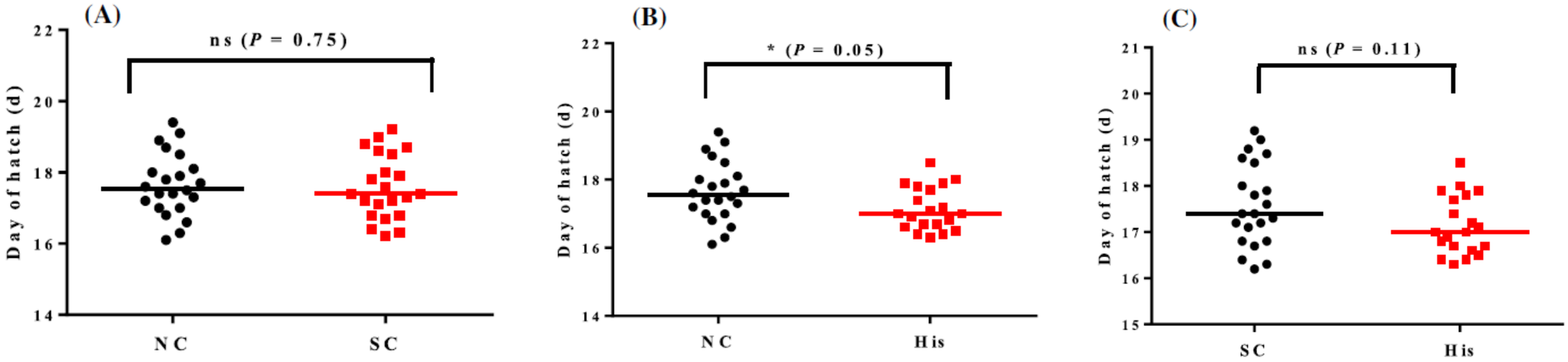


→ Histidine is not only used for protein synthesis — it is also a precursor of muscle dipeptides.

Haug et al., 2008

Histidine Improves Hatch Performance

- ***In ovo*** **injection of 1% concentrated histidine solution at 13 D of incubation might have beneficial effects on hatching activities**



NC: Non-injected control group
SC: 0.75% Saline control group
His: 1% Histidine (compared to total histidine in the egg) solution injected group

Xu et al., 2019, Poultry Science

Take Home Message

1 HIS, the bottleneck in low-CP diets

Performance follows only if His is balanced — **dHis:dLys 37–39%**, raised again in NRC 2026.

2 Disease challenge raises the requirement

Gut damage cuts absorption, immunity raises demand — peak in the acute phase. Breast meat needs more than growth: **0.37-0.49% SID His**.

3 HIS is more than a growth amino acid

His increased **carnosine in breast muscle up to +64% — pH buffering and antioxidant capacity (T-AOC↑, MDA↓, GPx↑)** deliver performance and meat quality.

In the Low-CP Era, It's Time to Rethink Histidine.

CJ BIO

Rethink Histidine, Achieve Precision.

Your Precision Nutrition Partner, CJ BIO